Math 5312 / Math 7363 Final Exam

Spring 2026

Student Name: _____

Submission Deadline: May 12, 2026.

Instructions

1. You must submit your solutions written by hand.
2. Typeset LaTeX submissions are **not** allowed.
3. You may write on paper or on a tablet/iPad using a stylus, but only handwritten work is permitted.
4. You are encouraged to discuss the problems with others, but each submitted solution must be written independently.
5. Show all essential steps of your reasoning.
6. Clearly state any theorem, lemma, proposition, or definition that you use.
7. Organize your work carefully and label each problem clearly.

Problem 1: Identity Theorem

State and prove the Identity Theorem for holomorphic functions on a connected domain.

Problem 2: Analytic Continuation

Let

$$f(z) = \sin z$$

and define

$$g(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}.$$

1. Using the Identity Theorem and Maclaurin series over $\mathbb{R}$, show that

$$f(z) = g(z) \quad \text{for all } z \in \mathbb{C}.$$

Problem 3: Open Mapping Theorem

State and prove the Open Mapping Theorem for nonconstant holomorphic functions on a domain.

Problem 4: Related Example for the Open Mapping Theorem

Consider the holomorphic function

$$f(z) = z^2.$$

Show that the image of the open disk

$$B_1(1) = \{z \in \mathbb{C} : |z - 1| < 1\}$$

under f is an open subset of $\mathbb{C}$.

Problem 5: Holomorphic Maps Between Compact Riemann Surfaces

State and prove the main proposition concerning a nonconstant holomorphic map

$$f : X \to Y$$

between compact Riemann surfaces, including the following properties:

- $f(X) = Y$,
- each fiber $f^{-1}(y)$ is finite,
- the set of ramification points is finite,
- away from branch values, f is a covering map,
- the sum of ramification indices over a fiber equals the degree.

Problem 6: Computation on the Torus to $\mathbb{CP}^1$

Let

$$T^2 = \mathbb{C}/\Lambda$$

be a complex torus.

Consider a nonconstant holomorphic map

$$f : T^2 \to \mathbb{CP}^1.$$

Using the main proposition, explain why such a map must be a branched covering map, and compute the degree of f in terms of the total ramification.

In particular, relate your answer to the fact that

$$\sum_{x \in f^{-1}(y)} e_x(f) = \deg f.$$

Problem 7: Fundamental Theorem of Algebra

State the Fundamental Theorem of Algebra and prove it using the proposition of Problem 5 for holomorphic maps between compact Riemann surfaces.

Problem 8: Residues of Meromorphic 1-Forms on a Compact Riemann Surface

Let M be a compact Riemann surface, and let

$$M = \bigcup_{\alpha} U_\alpha$$

be an open cover by coordinate charts. Suppose ω is a nonzero meromorphic 1-form on M, so that on each chart U_α with local coordinate z_α, one can write

$$\omega = f_\alpha(z_\alpha)\, dz_\alpha,$$

where f_α is meromorphic in the coordinate z_α.

Let $p \in M$ be a pole of ω. Choose a chart $U_\alpha \ni p$ with local coordinate z_α such that

$$z_\alpha(p) = 0,$$

and write the Laurent expansion

$$f_\alpha(z_\alpha) = \sum_{n=-m}^{\infty} a_n z_\alpha^n.$$

Define the residue of ω at p by

$$\operatorname{Res}_p(\omega) := a_{-1}.$$

Prove the following statements.

1. The number $\operatorname{Res}_p(\omega)$ is well defined; that is, it is independent of the choice of coordinate chart and local coordinate used to compute it.

2. If $p_1, \ldots, p_n$ are the poles of ω, then

$$\sum_{j=1}^{n} \operatorname{Res}_{p_j}(\omega) = 0.$$

Your proof of the second statement must use the generalized Stokes theorem on

$$M \setminus \bigcup_{j=1}^{n} B_j,$$

where each B_j is a sufficiently small coordinate disk around the pole p_j.

Problem 9: Residue Theorem and the Degree of a Principal Divisor

Let M be a compact Riemann surface, and let f be a nonzero meromorphic function on M.

Define the principal divisor

$$(f) = \sum_{p \in M} \nu_p(f)\, p.$$

State and prove that

$$\deg(f) = 0.$$

Your proof must explicitly proceed by considering the meromorphic 1-form

$$\omega = \frac{df}{f},$$

showing first that

$$\sum_{p \in M} \mathrm{Res}_p\left(\frac{df}{f}\right) = 0,$$

and then deducing from the local computation of the residue of df/f that

$$\deg(f) = \sum_{p \in M} \nu_p(f) = 0.$$

Problem 10: Principal Divisor on $\mathbb{CP}^1$ and the Canonical Divisor of dz

Consider the meromorphic function on $\mathbb{CP}^1$

$$f(z) = \frac{z^2(z-1)}{(z-2)^3}.$$

Compute the principal divisor (f) on $\mathbb{CP}^1$, including the contribution at ∞, and verify directly that

$$\deg(f) = 0.$$

Then consider the meromorphic 1-form dz on $\mathbb{C} \subset \mathbb{CP}^1$. Extend it meromorphically to $\mathbb{CP}^1$, compute its divisor on $\mathbb{CP}^1$, and show that its degree is

$$\deg(dz) = -2.$$

Problem 11: Riemann–Hurwitz Theorem

State and prove the Riemann–Hurwitz theorem for a nonconstant holomorphic map

$$f : X \to Y$$

between compact Riemann surfaces.

Your statement should include the degree $n = \deg f$, the ramification points, their ramification indices, and the Euler characteristics of X and Y.

Problem 12: Degree of a Canonical Divisor

Let M be a compact Riemann surface, and let ω be a nonzero meromorphic 1-form on M. Its divisor is called a canonical divisor.

Using the Riemann–Hurwitz theorem, prove that

$$\deg(\omega) = -\chi(M).$$

Problem 13: The Spaces $L(D)$, $I(D)$, and the Case $L(p)$

Let X be a compact Riemann surface, and let D be a divisor on X.

Recall the definitions

$$L(D) := \{f \text{ meromorphic on } X : (f) + D \geq 0\} \cup \{0\},$$

and

$$I(D) := \{\omega \text{ meromorphic 1-forms on } X : (\omega) - D \geq 0\} \cup \{0\}.$$

Let K be a canonical divisor on X.

Prove the following statements:

1. There is a natural vector space isomorphism

$$I(K - D) \cong L(D).$$

2. There is a natural vector space isomorphism

$$I(D) \cong L(K - D).$$

3. If $p \in X$, then

$$\ell(p) := \dim_{\mathbb{C}} L(p) \leq 2.$$

Problem 14: Riemann–Roch, Step 1

Let X be a compact Riemann surface of genus g, and let $D \geq 0$ be an effective divisor on X.

State the following criterion for principal parts:

Given distinct points $a_1, \ldots, a_n \in X$ and principal parts $f_1, \ldots, f_n$ prescribed at those points, the following are equivalent:

1. There exists a meromorphic function on X having principal part f_j at each a_j and having no other poles.
2. For every holomorphic 1-form ω on X,
$$\sum_{j=1}^{n} \operatorname{Res}_{a_j}(f_j \omega) = 0.$$

Using this criterion, prove that if $D \geq 0$, then

$$\ell(D) - i(D) = \deg D + 1 - g.$$

Problem 15: Riemann–Roch, Step 2

Let X be a compact Riemann surface of genus g, and let D be an arbitrary divisor on X.

Prove that

$$\ell(D) - i(D) \geq \deg D + 1 - g.$$

Your proof should proceed by comparing D and $D - a$ for a point $a \in X$, and by ruling out the possibility that both

$$\ell(D - a) = \ell(D) - 1 \qquad \text{and} \qquad i(D - a) = i(D) + 1$$

occur simultaneously.

Problem 16: Riemann–Roch, Step 3 and Final Conclusion

Let X be a compact Riemann surface of genus g, and let D be any divisor on X.

Using the result of Problem 15 applied to the divisor $K - D$, together with the identification

$$I(D) \cong L(K - D),$$

prove the reverse inequality

$$\ell(D) - i(D) \leq \deg D + 1 - g.$$

Deduce that

$$\ell(D) - i(D) = \deg D + 1 - g.$$

This is the Riemann–Roch theorem.

Problem 17: The Case $\ell(p) \geq 2$ and the Sphere

Let X be a compact Riemann surface.

Prove that the following are equivalent:

1. X is homeomorphic to S^2.
2. There exists a point $p \in X$ such that
$$\ell(p) = \dim_{\mathbb{C}} L(p) \geq 2.$$
3. X is biholomorphic to $\mathbb{CP}^1$.

Problem 18: Asymptotic Riemann–Roch and Base Point Freeness

State the asymptotic Riemann–Roch theorem for a divisor D on a compact Riemann surface X, and use it to prove the following statement:

If

$$\deg D \geq 2g,$$

then for every point $p \in X$, there exists a function

$$f \in L(D)$$

such that

$$f(p) \neq 0.$$

In other words, prove that the complete linear system $L(D)$ is base point free.

Problem 19: Separation of Points

Let X be a compact Riemann surface of genus g, and let D be a divisor on X such that

$$\deg D \geq 2g + 1.$$

Prove that $L(D)$ separates points; that is, for any two distinct points $p, q \in X$, there exists a function

$$f \in L(D)$$

such that

$$f(p) \neq f(q).$$

Problem 20: Separation of Tangent Directions and Immersions

Let X be a compact Riemann surface of genus g, and let D be a divisor on X such that

$$\deg D \geq 2g + 1.$$

Recall that a holomorphic map

$$\Phi : X \to \mathbb{CP}^N$$

is called an immersion at a point $p \in X$ if its differential

$$d\Phi_p : T_pX \to T_{\Phi(p)}\mathbb{CP}^N$$

is nonzero. If this holds for every $p \in X$, then Φ is called an immersion.

Prove that $L(D)$ separates tangent directions in the following sense:
for every $p \in X$, there exist functions

$$f_0, f_1 \in L(D)$$

such that

$$f_0(p) \neq 0 \qquad \text{and} \qquad \operatorname{ord}_p(f_1) = 1.$$

Note: Why the Kodaira Map is an Embedding, and Why a Compact Riemann Surface is an Algebraic Curve

Let X be a compact Riemann surface of genus g, and let D be a divisor on X such that

$$\deg D \geq 2g + 1.$$

Choose a basis

$$s_0, \ldots, s_N$$

of $L(D)$, and define the Kodaira map

$$\Phi_D : X \to \mathbb{CP}^N, \qquad x \longmapsto [s_0(x) : \cdots : s_N(x)].$$

The previous problems together show the following:

1. By base point freeness, for every $x \in X$, at least one basis element $s_j(x)$ is nonzero. Hence Φ_D is well defined.
2. By separation of points, if $p \neq q$ in X, then there exists $f \in L(D)$ such that $f(p) \neq f(q)$. Therefore $\Phi_D(p) \neq \Phi_D(q)$, so Φ_D is injective.
3. By separation of tangent directions, the differential of Φ_D is nonzero at every point. Hence Φ_D is an immersion.
4. Since X is compact and $\mathbb{CP}^N$ is Hausdorff, a continuous injective map

   $$\Phi_D : X \to \Phi_D(X)$$

   is a homeomorphism onto its image.

Therefore Φ_D is a holomorphic embedding of X into projective space.

Now recall Chow's theorem:

> Every closed complex submanifold of projective space $\mathbb{CP}^N$ is a projective algebraic variety.

Since $\Phi_D(X) \subset \mathbb{CP}^N$ is the image of a compact Riemann surface under a holomorphic embedding, it is a closed complex submanifold of $\mathbb{CP}^N$. By Chow's theorem, $\Phi_D(X)$ is a projective algebraic curve.

Because X is biholomorphic to $\Phi_D(X)$, we conclude that X itself is biholomorphic to a projective algebraic curve.

Hence the arguments above establish the conclusion:

Every compact Riemann surface is an algebraic curve.

Problem 21: A Double Pole on a Complex Torus

Let

$$T = \mathbb{C}/\Lambda$$

be a complex torus, where $\Lambda \subset \mathbb{C}$ is a lattice, and let $p \in T$.

Prove that there exists a nonconstant meromorphic function

$$f \in \mathcal{M}(T)$$

having a pole at p of order at most 2.

Then prove that if T is not biholomorphic to $\mathbb{CP}^1$, one may choose such an f so that the pole at p has order exactly 2.

Problem 22: Minimal Polynomial of $\wp'$

Let $\wp$ be the Weierstrass function with Laurent expansion

$$\wp(z) = \frac{1}{z^2} + a_2 z^2 + a_4 z^4 + O(z^6).$$

Show that $\wp'$ satisfies

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$

for suitable constants g_2, g_3, and deduce that the minimal polynomial of $\wp'$ over $\mathbb{C}(\wp)$ is

$$T^2 - \big(4\wp^3 - g_2\wp - g_3\big).$$

Note: Elliptic Integrals, Riemann's Original Motivation, and the Torus

One of the original motivations for Riemann's point of view was the study of integrals of algebraic functions. A basic example from elementary calculus is the inverse sine function:

$$\arcsin(b) - \arcsin(a) = \int_a^b \frac{dx}{\sqrt{1-x^2}}.$$

At first sight this already looks more complicated than the integral of a rational function, because of the square root in the denominator. However, after introducing the substitution $x = \sin\theta$, the integral becomes

$$\int_a^b \frac{dx}{\sqrt{1-x^2}} = \int_{\arcsin(a)}^{\arcsin(b)} d\theta,$$

so in this case the multivalued algebraic expression can still be reduced to an elementary inverse trigonometric function.

Elliptic integrals arise when one considers similar expressions but with a cubic or quartic polynomial under the square root, for example

$$\int_a^b \frac{dx}{\sqrt{P_3(x)}} \qquad \text{or} \qquad \int_a^b \frac{dx}{\sqrt{P_4(x)}},$$

where P_3 and P_4 are polynomials of degree 3 or 4 with distinct roots. These integrals appear naturally in classical geometry, mechanics, and celestial mechanics. In particular, problems involving the motion of a body along a curved trajectory, after a suitable change of variables, often lead to integrals of this type.

The crucial difference from the inverse sine example is that, in general, such integrals cannot be reduced to elementary functions. Riemann's insight was that one should not regard

$$\frac{dx}{\sqrt{P_3(x)}}$$

or

$$\frac{dx}{\sqrt{P_4(x)}}$$

as an ill-behaved multivalued expression on the complex x-plane. Instead, one should enlarge the domain and pass to the algebraic curve

$$y^2 = P_3(x) \qquad \text{or} \qquad y^2 = P_4(x).$$

On this curve, the expression

$$\frac{dx}{y}$$

is a genuine meromorphic, and in the nonsingular case holomorphic, 1-form. Thus the original integral is reinterpreted as an integral of a differential on a Riemann surface.

So the conceptual shift is this:

integral of an algebraic expression in x $\longrightarrow$ integral of a meromorphic 1-form on a Riemann surface.

In the genus 0 case, the corresponding Riemann surface is the Riemann sphere, and one recovers classical inverse trigonometric or logarithmic functions. The inverse sine integral is the model example: it comes from a double cover of the x-sphere branched at ± 1, but this surface is still genus 0, so the integral remains elementary.

In the elliptic case, however, the curve

$$y^2 = 4x^3 - g_2 x - g_3$$

defines a compact Riemann surface of genus 1. Such a surface is biholomorphic to a complex torus

$$\mathbb{C}/\Lambda.$$

This is where the Weierstrass functions enter.

Let $\Lambda \subset \mathbb{C}$ be a lattice and consider the torus

$$\mathbb{C}/\Lambda.$$

The Weierstrass functions $\wp$ and $\wp'$ satisfy the algebraic relation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Hence the torus can be realized as the algebraic curve

$$y^2 = 4x^3 - g_2 x - g_3.$$

If we write

$$w = \wp(z),$$

then

$$dw = \wp'(z)\, dz,$$

so

$$dz = \frac{dw}{\sqrt{4w^3 - g_2 w - g_3}}.$$

Therefore the elliptic integral

$$\int_a^b \frac{dw}{\sqrt{4w^3 - g_2 w - g_3}}$$

is nothing but the integral of the holomorphic differential dz after passing from the algebraic curve to its uniformization by the torus.

This explains why the inverse of an elliptic integral is not single-valued on the complex plane: it becomes single-valued only after lifting to the universal cover $\mathbb{C}$, where the ambiguity is measured by the lattice of periods Λ. In other words, the multivaluedness of the inverse is precisely the shadow of the torus structure.

Thus Riemann's original motivation can be summarized as follows:

- elementary inverse functions such as arcsin arise from genus 0 situations;
- elliptic integrals arise from genus 1 algebraic curves;
- instead of integrating multivalued algebraic expressions on the plane, one should integrate meromorphic or holomorphic differentials on the appropriate Riemann surface.